

Vraj Dobariya

+91 9898760222 | vrajdobariya.ict18@gmail.com | Surat , India | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)
Gold Medalist in Computer Science | GATE Top 15% - Competitive National Exam in CS | AI/Data Enthusiast

Domain/Functional Areas

- Healthcare
- Recommender Systems
- Gen AI to Agentic AI
- Risk Management & Fraud Detection

Key Skills

Programming Languages: Python, R, SQL, C, C++

Databases & Vector Stores: MySQL, MongoDB, ChromaDB(Vector Search)

AI/ML Frameworks & Libraries: Scikit-learn, XGBoost, PyTorch, TensorFlow, Keras, Statsmodels, SciPy (stats), Prophet, pmdarima, OpenCV, spaCy, NLTK, Transformers, Hugging Face, LangChain, LangGraph, FastAPI, Flask API, Streamlit, CUDA

MLOps & Deployment: Docker, Kubernetes, CI/CD Pipelines, AWS, MLflow, Linux, Unix-based OS, Git

AI/ML Expertise: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Generative AI, LLM, RAG, Agentic Workflows, Model Fine Tuning(Transfer Learning), Model Evaluation, Prompt Engineering, Reinforcement Learning

Data Science & Analytics: EDA(Numpy, Pandas, Matplotlib, Seaborn), Data Preprocessing, Feature Engineering, Hypothesis Testing, A/B Testing, Time Series Analysis, Forecasting, Big Data Technologies(Apache Spark, Hadoop), Data Visualization (Tableau, Power BI)

Core CS: Data Structures & Algorithms, Graph Analysis, Software Engineering (System Design, Testing, OOP), API Integration (REST, GraphQL), Agile Methodologies

Professional Experience

AtliQ Technologies Pvt. Ltd.

Vadodara, India

AI/Data Intern (Remote) (Internship)

Dec 2024 - May 2025

- Worked with a cross-functional team engaging stakeholders across healthcare and generative AI domains to align solutions with business goals, driving communication that expedited delivery and improved domain impact.
- Built a hybrid BioBERT + Logistic Regression model for medical transcription, achieving 79.5% accuracy via SMOTE and SynonymAug; reduced manual effort by 50%, cut misclassifications by 25%, and ensured reliable, scalable EHR support.
- Built an intelligent chatbot using RAG with LLaMA 3.3 and ChromaDB, increasing user engagement 60% and revenue potential 40%, integrating semantic routing and real-time SQL queries to optimize response times and experience.

Skills: BioBERT / Logistic Regression / SMOTE / RAG / LLaMA (LLMs) / ChromaDB / SQL / Flask API / Docker / AWS EC2 / Teamwork

SAMP Technologies LLP

Surat, India

Data Scientist (Full-Time)

Jan 2022 - Jan 2024

- Collaborated with business stakeholders and cross-functional teams to identify analytical opportunities, translate requirements into scalable data science solutions, and drive data-informed decision-making to improve operational efficiency.
- Developed a Logistic Regression credit risk model, achieving 94% recall using SMOTE-Tomek and Optuna; reduced false negatives by 15%, generated scalable credit scores, and deployed a FastAPI service with Docker.
- Built a K-Means model on 1M+ transactions using RFM analysis, boosting campaign precision by 35%; engineered 6 scaled features with IQR capping and PCA, reducing design time by 40% and improving interpretability by 25% with SQL/SAS.

Skills: Scikit-learn / K-Means / XGBoost / SMOTE / Optuna / FastAPI / Docker / SQL / A/B Testing / T-Test / ANOVA / Teamwork

Projects

Personalized Product Recommendation System - E-commerce and Recommender Systems [Project Link](#)

[Python / TensorFlow / Keras / Tokenization / LSTM / Bidirectional RNN / Streamlit]

Nov 2024 - Jan 2025

- Developed an LSTM and Bidirectional LSTM recommender system on Retailrocket data, achieving 85.35% Precision@5 and Recall@5 with dual item-category inputs, improving personalization by 20% over baselines.
- Optimized model with robust preprocessing, IQR outlier detection, tokenization, and a Streamlit app with caching, enabling interactive visualizations and enhancing user engagement and recommendation efficiency.

RAG-Driven Research Tool - Real Estate [Project Link](#)

[Langchain / Transformers / ChromaDB / FastAPI / System Design / Communication / Agentic Workflows]

Aug 2024 - Oct 2024

- Engineered a Streamlit-based web app with Retrieval-Augmented Generation (RAG), slashing LLM API costs by ~70% and research time by ~50% for real estate insights, delivering precise, URL-driven answers using LangChain's UnstructuredURLLoader tool.
- Spearheaded the integration of HuggingFace's all-MiniLM-L6-v2 embeddings and ChromaDB retrieval system, driving analytical efficiency and fostering teamwork to produce scalable, source-referenced outputs via Llama3 platform.

Education

California State University, East Bay

Hayward, USA

Master of Science - Statistics Data Science

Jan 2024 - May 2026

Adani University

Ahmedabad, India

Bachelor of Technology - Information Technology

Jun 2018 - Jul 2022